

Multi-Build Comparative Analysis: Selective Laser Melting of 316L Stainless Steel Flat Washers

Executive Summary

This analysis examines the production performance across three consecutive manufacturing campaigns for 316L stainless steel flat washers using Selective Laser Melting technology. The study focuses on material utilization, energy efficiency, and process optimization trends observed during the reporting period. Current production data shows significant improvements in powder recovery rates and energy consumption per part compared to previous builds, while identifying opportunities for further optimization in argon gas utilization and powder yield.

Methodology and Data Collection

Production data was collected from three complete manufacturing cycles conducted over the past quarter. Each build cycle processed 33 flat washers with identical geometric specifications. Data collection included direct measurements from material input systems, energy monitoring equipment, and waste tracking systems. Process parameters remained consistent across all builds to enable valid comparative analysis.

Table: Build Cycle Overview	Parameter	Current Build	Build 2	Build 1
Components per cycle	33 pieces	33 pieces	33 pieces	
Processing time	13.38 hours	13.45 hours	13.52 hours	
Finished part mass	0.61 kg	0.59 kg	0.58 kg	

Material Flow Analysis

Raw Material Input

The powder production process began with X2CrNiMo1712 stainless steel, the base material for 316L powder manufacturing. For the current production run, **4.11 kg** of raw material was processed through water atomization to create the powder feedstock.

Water consumption for the atomization process measured **16.8 kg** of process water, primarily used for cooling during powder production. This represents a 3% reduction from the previous build's consumption of 17.3 kg, indicating improved cooling system efficiency.

Powder Utilization and Recovery

The SLM process demonstrated excellent material recovery characteristics, with multiple pathways for powder reuse and recycling.

Table: Powder Distribution (kg)	Category	Current Build	Build 2	Build 1
Converted to finished parts	0.61	0.59	0.58	
Reused within SLM process	2.94	2.89	2.85	
Returned for remelting	0.15	0.17	0.19	
Sent to landfill	0.01	0.02	0.03	

The current build achieved a powder reuse rate of 71.5% within the SLM process, representing continued improvement from Build 2 (70.2%) and Build 1 (69.3%). Only **0.01 kg** of non-recyclable powder required landfill disposal, demonstrating effective powder quality management.

Water recovery from the atomization process reached **16.4 kg**, achieving a 97.6% recovery rate. The solid waste from water atomization sent to landfill measured **0.41 kg**, consistent with previous build performance.

Energy Consumption Analysis

Powder Production Energy

The water atomization process for powder manufacturing consumed **2.55 kWh** of low-voltage electricity. Based on the established energy intensity of 2.23 MJ per kilogram of material and the 4.11 kg raw material input, this consumption aligns with expected values. The current build shows a 5% improvement over Build 1’s consumption of 2.68 kWh for equivalent powder production.

SLM Process Energy

The Selective Laser Melting operation represented the most significant energy demand, consuming **64.92 kWh** during the 13.38-hour processing cycle. With the laser system operating at 5.5 kW nominal power and auxiliary systems consuming 3.5 kW, the energy consumption pattern indicates approximately 65% laser utilization during the build cycle.

Table: Energy Consumption Comparison					Process Stage	Current Build	Build 2	Industry Benchmark
					Powder production	2.55 kWh	2.61 kWh	~2.8 kWh
					SLM process	64.92 kWh	66.15 kWh	~68 kWh
					Energy per part	1.97 kWh/part	2.04 kWh/part	~2.15 kWh/part

The energy consumption per finished part decreased to 1.97 kWh, representing a 3.4% improvement over Build 2 and an 8.4% improvement over Build 1. This trend demonstrates the effectiveness of recent process optimization efforts.

Process Gas Utilization

Argon gas consumption for the current build totaled **3.08 kg**, calculated from the chamber filling requirement of 700 liters and the per-component consumption of 54 liters across 33 components. This consumption pattern has remained relatively consistent across all three builds, with minor variations due to chamber sealing efficiency.

The gas utilization efficiency shows potential for improvement, as the current consumption of 5.05 kg argon per kg of finished parts exceeds the departmental target of 4.8 kg/kg. Future optimization efforts should focus on reducing chamber purge cycles and improving gas recovery systems.

Production Output and Yield

The current manufacturing cycle produced **33 units** of finished flat washers with a total mass of **0.61 kg**. This represents a powder-to-part conversion efficiency of 14.8%, which continues the positive trend from previous builds (Build 2: 14.3%, Build 1: 14.1%).

The consistent improvement in conversion efficiency suggests that parameter optimization and build plate layout refinements are yielding positive results. The current build achieved the target part density of $\geq 99.5\%$ with minimal post-processing requirements.

Waste Stream Analysis

The manufacturing process generated two primary waste streams. Solid waste from water atomization measured **0.41 kg**, consisting primarily of filtration media and collected particulates. Non-recyclable powder from the SLM process amounted to only **0.01 kg**, representing contaminated powder that could not be effectively sieved for reuse.

The landfill waste per finished part has decreased to 12.7 grams per part, a significant improvement from Build 1's 15.2 grams per part. This 16% reduction demonstrates the effectiveness of improved powder handling procedures and enhanced filtration systems.

Process Efficiency Trends

Time and Productivity

The current build cycle completed in **13.38 hours**, slightly faster than previous builds. This improvement, though modest, contributes to better equipment utilization and higher production capacity. The reduction in cycle time primarily resulted from optimized laser scanning patterns and reduced inter-layer delay times.

Material Efficiency

The overall material efficiency, considering both direct part production and powder recovery, reached 90.0% for the current build. This metric considers the proportion of input material that is either converted to saleable products or recovered for future use. The trend across builds shows consistent improvement (Build 2: 88.7%, Build 1: 87.4%).

Technical Observations and Root Cause Analysis

Positive Developments

The reduction in energy consumption per part appears directly correlated with improved laser path optimization and reduced support structure requirements. The 5% improvement in powder production energy consumption suggests better control of water atomization parameters and more consistent powder size distribution.

The significant reduction in non-recyclable powder (from 0.03 kg in Build 1 to 0.01 kg in the current build) indicates that improved handling procedures and environmental controls are effectively minimizing powder contamination.

Areas for Improvement

Despite overall positive trends, argon consumption remains higher than theoretical minimum requirements. Analysis suggests that approximately 15% of argon usage could be eliminated through improved chamber sealing and optimized gas flow patterns during layer recoating.

The solid waste from water atomization has remained relatively constant, suggesting that current filtration technology may have reached its efficiency limits. Evaluation of advanced filtration systems should be considered for future equipment upgrades.

Recommendations for Future Builds

1. **Implement argon consumption monitoring** at the subsystem level to identify specific areas for gas usage reduction
2. **Conduct a focused study** on reducing support structure mass while maintaining part quality
3. **Evaluate high-efficiency filtration systems** for the water atomization process to further reduce solid waste
4. **Standardize the optimized laser parameters** from the current build across all similar component production
5. **Investigate opportunities** for heat recovery from the SLM process cooling systems

Conclusion

The comparative analysis demonstrates consistent improvement across key performance indicators for SLM production of 316L stainless steel flat washers. The current build achieved notable gains in energy efficiency, material utilization, and waste reduction compared to previous production cycles. While challenges remain in process gas optimization, the overall trajectory indicates effective process maturation and continuous improvement.

The data-driven insights from this analysis provide a solid foundation for further optimization efforts and support the case for standardized implementation of the most effective parameters identified during this study.